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Course: Calculus Beta

Assignment: MyLab Math - Uge 10

Solve the differential equation $(x-2)\frac{dy}{dx} + y = 3e^x$, $x > 2$.

To solve the linear equation, $\frac{dy}{dx} + P(x)y = Q(x)$, multiply both sides of the equation by the integrating factor $e^{\int P(x)dx}$ and integrate both sides of the equation.

First, put the equation into standard form, $\frac{dy}{dx} + P(x)y = Q(x)$.

To put the given equation into standard form, divide by $x-2$ to obtain $\frac{dy}{dx} + \frac{1}{x-2}y = \frac{3}{x-2}e^x$, giving $P(x) = \frac{1}{x-2}$.

Now find the integrating factor, $e^{\int P(x)dx}$. Use a constant of integration $C = 0$, so y is as simple as possible. Note that since $x > 2$, the absolute value signs can be omitted.

$$\begin{aligned} e^{\int P(x)dx} &= \int \frac{1}{x-2} dx \\ &= e^{\ln(x-2)} \\ &= x-2 \end{aligned}$$

Multiply the equation in standard form by the integrating factor, $x-2$.

$$\begin{aligned} (x-2)\left(\frac{dy}{dx} + \frac{1}{x-2}y\right) &= (x-2)\left(\frac{3}{x-2}e^x\right) \\ (x-2)\frac{dy}{dx} + y &= 3e^x \end{aligned}$$

The integrating factor was chosen such that the left side of the equation, $(x-2)\frac{dy}{dx} + y$, is the derivative of the product $(x-2)y$.

$$\begin{aligned} \frac{d}{dx}((x-2)y) &= 3e^x && \text{Rewrite the left hand side of the equation.} \\ (x-2)y &= 3\int e^x dx && \text{Integrate both sides of the equation.} \\ &= 3e^x + C && \text{Simplify.} \end{aligned}$$

Finally, divide both sides of the equation by $x-2$ to obtain $y = \frac{1}{x-2}(3e^x + C)$.

Therefore, the solution of $(x-2)\frac{dy}{dx} + y = 3e^x$ is $y = \frac{1}{x-2}(3e^x + C)$.